Linearized Amplifier Technologies and Services Pvt. Ltd.

A deep-tech startup incubated at I.I.T Roorkee

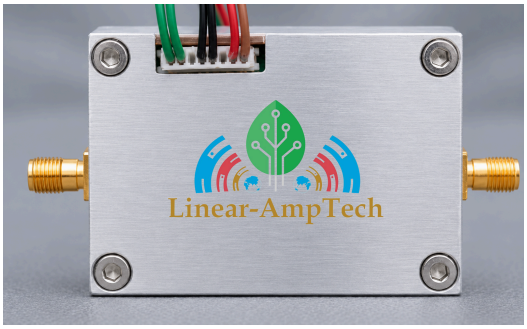
LAMP-PACAB-0.4-0.52-50W

UHF-Band RF Power Amplifier | 400–520 MHz

The LAMP-PACAB-0.4-0.52-50W is a compact, high-efficiency solid-state power amplifier module designed for continuous coverage across the 400–520 MHz UHF band. Packaged in a rugged enclosure, the power amplifier module delivers 45–54 W of output power with drain efficiency up to 58.9%, making it ideal for UHF range wireless applications.

KEY FEATURES

- Frequency band: 400-520 MHz
- Output power: 46.5–47.4 dBm
- Drain efficiency: up to 58.9% at saturation
- Gain flatness: ±0.5 dB across the operating band
- Harmonic suppression: better than 41 dBc
- spurious emissions: better than 77 dBc
- RF ramp up/down: < 12.3 µS
- Compact, rugged form factor



LAMP-PACAB-0.4-0.52-50W module (actual appearance may vary)

PERFORMANCE SPECIFICATIONS (TESTED AND VALIDATED)

Parameter	Conditions	Value
Frequency range	—	400–520 MHz
Output power	Pin = 10 dBm, CW	46.5–47.4 dBm
Power gain	Pin = 10 dBm	36.5–37.4 dB
Gain flatness	Pin = 10 dBm	±0.3–±0.5 dB
Drain efficiency	Pin = 10 dBm	51.3–58.9 %
2nd harmonic suppression	At Psat	< −41 dBc
3rd harmonic suppression	At Psat	< −66 dBc
Spurious emissions	At Psat	≤ −77 dBc
C/IMD3 (two-tone)	6 dB back-off from Psat, 1 MHz tone spacing	< −25.7 dBc
RF ramp up/down time	10 kHz square-wave envelope	< 12.3 µS

ELECTRICAL & MECHANICAL SPECIFICATIONS

Supply voltage (main)	28 VDC
Supply voltage (control)	5 VDC
Quiescent current (main supply)	1.8–1.9 A
Quiescent current (control supply)	0.05 A
Operating current	~3.5 A @ Pin = 10 dBm
Enclosure dimensions	72 mm × 50.8 mm × 16.6 mm
RF connectors	SMA

DC SUPPLY

Pin	Function	Specification
1, 2	VDD	+28 VDC
4, 5	GND	Ground
7	Switch ON/OFF	Enable: 3–5 V | Disable: 0–2.5 V | 50 mA consumption
8	Temperature monitor	Reference voltage 2.85 V @ 25°C

TYPICAL APPLICATIONS

- Tactical Communications & Military Radios
- Aerospace & Avionics Systems
- Broadcast and telemetry links in the 400–520 MHz band
- SATCOM and beyond-line-of-sight communication front-ends
- Electronic Warfare (EW) & Radar Systems

ORDERING INFORMATION

Part number	LAMP-PACAB-0.4-0.52-50W
Description	UHF-Band (400–520 MHz) RF Power Amplifier Module
Standard kit includes	PA module, 8-Pins Interface housing
Contact	sales@linearamptech.com | www.linear-amptech.com

Specifications subject to change without notice. Data based on measured results from prototype validation (May 2026). Please contact Linear-AmpTech for application-specific guidance.